

Nishika Panchal

Data & Backend Engineer · Snowflake · dbt · AWS · Java

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PROFESSIONAL PROFILE

Data & Backend Engineer with 6+ years of experience designing scalable data platforms and cloud-native backend systems. Specialised in building end-to-end ELT pipelines using Snowflake and dbt Core, with hands-on experience in data modelling, data quality, and AWS cloud integration. Background in Java, Spring Boot, and event-driven architectures (Kafka) provides strong technical depth across the full data stack. Collaborative team player with a strong ownership mindset and a focus on delivering reliable, analytics-ready data to business stakeholders.

PROFESSIONAL EXPERIENCE

Career Break – Upskilling & Certification Preparation | Self-Directed

Apr 2026 – Present

- Preparing for SnowPro Core Certification (Snowflake) — deepening expertise in Snowflake architecture, performance optimisation, data sharing, and security
- Preparing for AWS Certified AI Practitioner certification — building foundational knowledge of AWS AI/ML services (SageMaker, Bedrock, Lambda)
- Improving German language skills to B2 level through active coursework and exam preparation

Software Engineer | IT care Holding AG, Germany

May 2025 – Mar 2026

- Designed and implemented a scalable cloud-native data warehousing platform on Snowflake Data Cloud, ingesting large-scale healthcare data (EHR, patient encounters, hospital billing) from AWS S3 data lake with IAM-enforced data governance
- Architected end-to-end ELT pipelines using dbt Core (CLI) following the Medallion Framework (Bronze → Silver → Gold) within Snowflake, transforming raw medical records into clean, analytics-ready datasets for BI teams
- Developed modular dbt Models with mixed materialisation strategies — Incremental models to process only new/changed records (reducing compute cost) and Ephemeral models as lightweight, reusable intermediate layers
- Built fully metadata-driven, parameterizable pipelines using advanced Jinja Templates and Macros, eliminating repetitive SQL logic and reducing code volume by ~40%
- Deployed dbt Snapshots (SCD Type 2) on critical patient health metrics and billing dimensions to maintain complete historical audit trails — preserving both old and new record states on every change
- Enforced production-grade data quality using automated dbt Singular Tests and schema assertions with severity flags (Warnings and Errors), blocking data anomalies such as duplicate Patient IDs and null clinical fields before deployment
- Applied Dimensional Modelling (Star Schema — Fact & Dimension tables) and One Big Table (OBT) patterns in the Gold layer, enabling BI teams to analyse patient readmission rates and operational KPIs
- Wrote Python scripts using boto3 to automate AWS S3 data ingestion workflows — handling file uploads, bucket triggers, and pre-processing before loading into the Snowflake data warehouse
- Configured Snowflake File Formats and Stages to ensure high-speed, cost-efficient loading of sensitive patient records into the data warehouse
- Integrated AWS cloud infrastructure (S3, IAM, EC2, CloudWatch) with Snowflake and managed multi-environment CI/CD deployments (Dev → QA → Prod) using Git branching and GitHub Actions
- Designed and operated real-time streaming ingestion pipelines alongside batch ELT workflows, using containerised microservices (Docker, Kubernetes) and event-driven messaging (Apache Kafka)

Senior Software Engineer | Volkswagen Group Technology Solutions, India

Aug 2022 – May 2023

- Implemented and extended backend features in distributed Java / Spring Boot architectures
- Worked closely with the DevOps team on Infrastructure-as-Code using Terraform to provision and manage cloud resources
- Performed root-cause analysis and technical debugging to improve system stability and performance
- Collaborated closely with cross-functional teams using Jira for agile project management

Senior Software Engineer | Espire Infolabs Pvt. Ltd., India

Mar 2021 – Jul 2022

Projects: *Conex & ResMed (CEVA Logistics Automation), PUC Capacity Planning Tool, Generic Pack App*

- Designed and developed scalable REST APIs and microservices with Java and Spring Boot for logistics and integration solutions
- Led migration from monolithic to microservices architecture, improving scalability and deployment frequency
- Drove architecture optimisations and code reviews; implemented unit tests (JUnit 5, Mockito) to ensure quality and performance
- Recognised with the Prime Member On-the-Spot Award for successful module integration

Software Engineer | [Reppify India Pvt. Ltd., India](#)

Mar 2018 – Mar 2021

Projects: *AccredAbility (MedTrainer), ICL Vehicle Information System*

- Implemented web services and OAuth 2.0 framework integration for accreditation and healthcare training systems
- Developed new features aligned with system design and requirements; acted as liaison to the client team
- Awarded Employee of the Month for consistent client delivery and successful OAuth 2 integration

German Integration Course – Level B1 | [VHS Leverkusen, Germany](#)

Feb 2024 – Feb 2025

TECHNICAL SKILLS

Languages	SQL, Python (boto3, Jinja, Snowpark), Java (11–21), JavaScript
Data Platform	Snowflake (Data Warehousing, Stages, File Formats, Streams, Tasks), Apache Kafka
Data Engineering	dbt Core (Medallion Architecture, Incremental/Ephemeral Models, Snapshots, SCD Type 2, Jinja Macros, Singular Tests), ETL/ELT, Star Schema, Dimensional Modelling, OBT
Cloud & IaC	AWS (S3, IAM, EC2, ECS, Lambda, CloudWatch, Bedrock), Docker, Kubernetes, Terraform (basics)
CI/CD & Ops	Git (branch workflow, multi-env deploy), GitHub Actions, GitLab CI, Jenkins, SonarQube, Monitoring & Observability
Backend	Spring Boot, Spring Framework, Spring Data JPA, Hibernate, Spring Security, OAuth2 / JWT, REST APIs, OpenAPI
Databases	Snowflake (Data Warehousing), PostgreSQL, MySQL, NoSQL (basics)
Certifications	SnowPro Core (in preparation), AWS Certified AI Practitioner (in preparation)
AI / ML	OpenAI API, LangChain, Prompt Engineering, AWS Bedrock (learning)
Tools	dbt Cloud, Git, Postman, IntelliJ IDEA, Jira, Bitbucket, Maven

EDUCATION

Master of Technology – Computer Science & Engineering

GGSIIP University, India

Sep 2015 – Jun 2018

Bachelor of Technology – Computer Science & Engineering

Maharshi Dayanand University, India

Apr 2010 – Jun 2014

LANGUAGES

German (B1, actively working towards B2) · **English** (Fluent) · **Hindi** (Native)